ND-120 FPGA Debug Reference Card

Basys3 (xc7a35tcpg236-1) • 100 MHz • 115200 Baud • Branch: redo-idb

LED Layout (left LD15 to right LD0)

TERM	CC3	CC2	CC1	CC0	—	—	—	LCS	MCLK	BEAT	UART	RST	RUN	GRN	RED
LD15	LD14	LD13	LD12	LD11	LD10	LD9	LD8	LD7	LD6	LD5	LD4	LD3	LD2	LD1	LD0

Switches (SW0=leftmost on Basys3)

Switch	Signal	UP (1)	DOWN (0)
SW0 (left)	sys_rst_n	**Reset released, CPU runs**	CPU held in reset
SW1	btn2	7-seg = MAC address	7-seg = MIC address

ND-120 CPU • Norsk Data 1988 • FPGA Implementation • 2026-03-30

LED Details

LED	Signal	ON	OFF
LD15	~TERM_n	**Cycle termination**	-
LD14-11	~CC3..0_n	**Cycle counter (cycles)**	-
LD7	~LCS_n	**Microcode loaded**	Loading
LD6	MCLK	**Memory clock**	No MCLK
LD5	clkTick[26]	**Blinks 1.5Hz**	No clock!
LD4	~uartTx	**TX active**	Silent
LD3	sys_rst_n	**Reset released**	In reset
LD2	~s_run	**CPU running**	OPCOM
LD1	cpu_led[1]	**GREEN**	-
LD0	cpu_led[0]	**RED (err)**	OK

7-Segment Display (4 hex digits)

SW1	Shows	Range
DOWN	MIC address (CSA_12_0, microcode)	0000-1FFF
UP	MAC address (LA_23_10, memory)	0000-3FFF

Changing values = CPU executing • Stuck 0000 = CPU not running

UART

115200	8N1	None	A17	A18
Baud	Format	Flow	TX pin	RX pin

Boot Sequence

1. Program bitstream
2. **LD5** blinks (clock running)
3. Slide SW0 UP (release reset)
4. **LD3** ON (reset released)
5. **LD15-11** cycling (state machine)
6. **LD7** ON after ~6ms (microcode loaded)
7. **LD2** ON (CPU running)
8. 7-seg shows changing hex
9. Terminal shows # prompt

Vivado ILA Debug Signals (mark_debug)

Signal	W	Description
s_debug_csa[12:0]	13	MIC address (microcode fetch)
s_debug_la_23_10[13:0]	14	MAC address upper (LA 23:10)
s_debug_ca_9_0[9:0]	10	MAC address lower (CA 9:0)
s_debug_cc_term[4:0]	5	{TERM,CC3,CC2,CC1,CC0}_n
s_debug_mclk	1	Memory clock
s_debug_lcs_n	1	LCS_n (0=loading, 1=loaded)
s_run	1	CPU run (0=running)
s_debug_uartTx	1	UART TX line
s_debug_uartRx	1	UART RX line
s_debug_cpu_led[6:0]	7	CPU status LEDs

Deep Hierarchy (add via Set Up Debug)

Path	W	Description
CPU_BOARD/.../s_FIDBO_15_0	16	CGA data bus output
CPU_BOARD/.../s_FIDBI_15_0	16	CGA data bus input
CPU_BOARD/CPU/CS/s_csbits	64	TOPCSB microcode word
CPU_BOARD/.../s_idb_erf_out	16	Register file to IDB
CPU_BOARD/.../CHIP_32H/txState	3	UART TX state machine
CPU_BOARD/.../CHIP_32H/rxState	3	UART RX state machine
CPU_BOARD/CYC/s_cc0_n	1	Cycle counter bit 0

Pin Map

Port	Pin	Port	Pin	Port	Pin
sysclk	W5	led[5]	U15	led[12]	P3
btn1/SW0	V17	led[6]	U14	led[13]	N3
btn2/SW1	V16	led[7]	V14	led[14]	P1
uartTx	A17	led[8]	V13	led[15]	L1
uartRx	A18	led[9]	V3	seg[0-6]	W7..U7
led[0]	U16	led[10]	W3	an[0-3]	U2..R2
led[1]	E19	led[11]	U3		
led[2]	U19				
led[3]	V19				